



# TextForge Studio

## Final Website Documentation

Multi-page AI writing, originality, content, SEO, student, business, social, and document tools platform

| Field                  | Details                                                                                                   |
|------------------------|-----------------------------------------------------------------------------------------------------------|
| Project Name           | TextForge Studio                                                                                          |
| Project Type           | Python Flask based professional text tools website                                                        |
| Final Build            | Bright professional multi-page version with payment, subscription, admin, AdSense, and hybrid API support |
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### Executive Summary

TextForge Studio is a professional web application for text rewriting, originality improvement, AI-style cleaning, SEO writing, student support, business writing, social media content, and document utilities.

The final website uses a clean light-blue and white gradient interface by default, includes a multi-page tools structure, and keeps dark mode as an optional user setting.

The platform includes signup/login, CAPTCHA, user plans, a 1-day trial, subscription workflow, manual Pakistani payment proof approval, configurable international payment links, admin panel, AdSense settings, and optional Gemini API mode.

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# 1. Project Overview

TextForge Studio is a full-featured text tools website built to help users generate, rewrite, polish, analyze, and export text through a clean, professional interface. It has been designed as a platform-neutral Python web app, so it can be hosted on any hosting provider that supports Python web applications.

The final build moves away from a crowded single-page dashboard and uses a multi-page workflow. Users first choose a tool category, then select a specific tool, and finally work inside a focused single-tool page. This improves usability, readability, and professional presentation.

## Main Purpose

Provide useful writing and text productivity tools for students, freelancers, marketers, business users, content creators, and general users.

Offer both built-in algorithm processing and optional Gemini API processing when a premium/trial user adds their own API key.

Support monetization through subscriptions, manual payments, international checkout links, and AdSense-ready ad placement areas.

# 2. What Was Made

The final website is a complete Python Flask application with frontend templates, static CSS/JavaScript, text processing backend, authentication, admin controls, plan management, payment proof workflow, AdSense settings, and export tools.

- A bright, professional multi-page interface with light-blue and white gradient styling by default.
- A top navigation system with Home, Guides, Pricing, Tools dropdown, About, Privacy, Terms, Disclaimer, Dashboard, and Admin access for admin users.
- A category-based tool library where each category has its own page and every tool opens in its own focused workspace.
- Signup and login with a simple CAPTCHA to reduce fake/bot accounts.
- User dashboard with plan status, daily usage, trial status, and subscription access.
- Admin-only dashboard for user management, payment proof approvals, subscription settings, ad settings, and payment method configuration.
- Hybrid processing: built-in algorithm mode and optional Gemini API mode using the user's own key.
- Import and export support for TXT, Markdown, DOCX, and PDF where supported by the installed Python libraries.
- AdSense-ready pages and configurable ad placement blocks controlled from the admin panel.

# 3. Final Website Structure

The website is organized into separate pages for better user experience and cleaner design. Instead of showing all tools, settings, inputs, outputs, and analytics on one screen, each part has a dedicated location.

| Route | Purpose                                        |
|-------|------------------------------------------------|
| /     | Homepage / landing page with overview and main |

|                       |                                                        |
|-----------------------|--------------------------------------------------------|
|                       | feature highlights                                     |
| /tools                | Main category index page showing all tool categories   |
| /tools/<category>     | Category page listing tools inside a selected category |
| /tool/<tool_id>       | Focused workspace for one selected text tool           |
| /dashboard            | User dashboard with plan and account status            |
| /pricing              | Pricing page for free, trial, and pro plans            |
| /subscribe            | Subscription/payment page with enabled payment options |
| /admin                | Admin-only control panel                               |
| /about                | About page with owner and project information          |
| /privacy-policy       | Privacy policy page                                    |
| /terms-and-conditions | Terms and conditions page                              |
| /disclaimer           | Responsible use and limitation disclaimer              |
| /guides               | SEO/support guide page for content value               |
| /ads.txt              | Dynamic AdSense ads.txt endpoint                       |
| /robots.txt           | Search engine crawling rules                           |
| /sitemap.xml          | Sitemap for search engine discovery                    |

## 4. User Roles and Access Levels

| Role            | Access                                                                                                                                        |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Guest           | Can view public pages such as home, pricing, tools overview, guides, policies, and legal pages.                                               |
| Registered User | Can login, use free tools within the configured daily character limit, and manage basic account activity.                                     |
| Trial User      | Receives a 1-day trial that unlocks higher limits and premium workflows during the trial period.                                              |
| Premium User    | Can use premium tools, higher character limits, export features, and Gemini API mode with their own key.                                      |
| Admin           | Has unlimited access, no feature limits, admin panel access, payment proof approval, user management, ad settings, and payment configuration. |

### Admin Login Details

Admin Email: muneebshah1192@gmail.com

Admin Password: Muneebshah1192@theadmin.com

For public deployment, change the admin password and SECRET\_KEY immediately before launch.

## 5. Public Website Pages

The public pages make the site more professional, trustworthy, and suitable for search engines and AdSense review. The website includes core business pages and policy pages.

| Page                 | Purpose                                                                                                                                                   |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Home                 | Explains the platform and directs users to tools, pricing, and dashboard.                                                                                 |
| Tools                | Shows tool categories in a clear structure and allows users to choose a category.                                                                         |
| Guides               | Adds useful text-writing and SEO-related content so the website is not only a tool dashboard.                                                             |
| Pricing              | Explains free, trial, and pro usage limits.                                                                                                               |
| About                | Includes creator details: Syed Muneeb, contact number, and email.                                                                                         |
| Privacy Policy       | Explains data handling, user account data, uploaded text, payment proof data, and user responsibilities.                                                  |
| Terms and Conditions | Defines acceptable use, subscription behavior, manual approval terms, and user responsibilities.                                                          |
| Disclaimer           | Clarifies that the platform assists with writing and originality support but cannot guarantee plagiarism-free, legal, academic, or professional outcomes. |

## 6. Tools System and Categories

The tool system is the main product area of TextForge Studio. Tools are grouped into categories and each category has a dedicated page. This makes the website cleaner and easier to understand for users of all age groups.

| Category               | Tool Count | Examples                                                                                                                                                                             |
|------------------------|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Originality & AI Tools | 7          | AI Plagiarism Remover, AI Humanizer, AI-Style Cleaner, AI Prompt Generator, Prompt Improver, Originality Risk Report, Similarity Checker                                             |
| Smart Writing          | 14         | AI Humanizer, Paragraph Rewriter, Deep Rewriter, AI-Style Cleaner, Sentence Shortener, Sentence Expander, Tone Converter, Professional Rewriter + 6 more                             |
| Content Intelligence   | 14         | Readability Analyzer, Writing Quality Score, Repetition Detector, Overused Word Detector, Weak Word Detector, Sentence Variety Score, Passive Voice Detector, Tone Analysis + 6 more |
| SEO & Blogging         | 14         | Blog Outline Generator, Blog Intro Generator, Blog Post Builder, Content Calendar, SEO Content Brief, SEO Meta Title, SEO Meta Description, URL Slug Generator + 6 more              |
| Student Tools          | 14         | Study Notes Generator, Important Points Extractor, MCQ Generator, Flashcard Creator, Definition Extractor, Glossary Builder, Thesis Statement Generator, Discussion Post             |

|                |    |                                                                                                                                                                                                      |
|----------------|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                |    | Writer + 6 more                                                                                                                                                                                      |
| Business Tools | 15 | Email Writer, Cold Email Writer, Proposal Writer, Meeting Notes Formatter, Meeting Agenda Builder, Customer Support Reply, Resume Bullet Improver, Cover Letter Builder + 7 more                     |
| Social Media   | 14 | Viral Hook Generator, YouTube Script Writer, Short Video Script, YouTube Title Generator, YouTube Description Writer, TikTok Hook Writer, Instagram Caption Generator, LinkedIn Post Writer + 6 more |
| Document Tools | 14 | Clean Formatting, Markdown Converter, Case Formatter, Bullet Point Generator, Outline Builder, Executive Summary, Pros & Cons Generator, Timeline Extractor + 6 more                                 |

## 6.1 Complete Category List

### Originality & AI Tools

- AI Plagiarism Remover: Rewrite text with stronger wording changes, sentence restructuring, and similarity-reduction support.
- AI Humanizer: Make robotic or stiff writing sound clearer, smoother, and more natural.
- AI-Style Cleaner: Reduce repetitive AI-style phrases and improve natural sentence flow.
- AI Prompt Generator: Create a detailed, structured prompt from a short idea or topic.
- Prompt Improver: Improve an existing prompt.
- Originality Risk Report: Estimate repeated wording and high-overlap risk using local text analysis.
- Similarity Checker: Compare the input and output text overlap.

### Smart Writing

- AI Humanizer: Rewrite robotic text into a more natural, human-style draft.
- Paragraph Rewriter: Rewrite paragraphs with changed wording and structure.
- Deep Rewriter: Aggressively restructures wording and sentence flow for a fresher draft.
- AI-Style Cleaner: Reduce robotic phrases, repeated transitions, and generic wording.
- Sentence Shortener: Make long writing concise.
- Sentence Expander: Add useful context and detail.
- Tone Converter: Convert text into professional, friendly, academic, or persuasive tone.
- Professional Rewriter: Make text polished and business-ready.
- Simple Rewriter: Make difficult text easy to understand.
- Active Voice Converter: Reduce passive wording where possible.
- Grammar & Punctuation Polish: Clean spacing, capitalization, and sentence punctuation.
- Clarity Enhancer: Improve flow, remove clutter, and strengthen wording.
- Transition Improver: Add smoother transitions between ideas.
- Sentence Variety Improver: Rewrite text with more varied sentence rhythm.

## Content Intelligence

- Readability Analyzer: Check reading level and writing stats.
- Writing Quality Score: Estimate clarity, structure, repetition, and readability.
- Repetition Detector: Find repeated words and phrases.
- Overused Word Detector: Find weak or repeated filler words.
- Weak Word Detector: Find vague, weak, or filler words.
- Sentence Variety Score: Analyze sentence length variation and rhythm.
- Passive Voice Detector: Find possible passive voice sentences.
- Tone Analysis: Analyze professional, emotional, urgent, and trust signals.
- Sentiment Analysis: Estimate positive/negative/neutral tone.
- Keyword Extractor: Extract important words and phrases.
- Content Brief Builder: Create a brief with audience, angle, keywords, and structure.
- Originality Risk Report: Estimate repeated wording and high-overlap risk using local text analysis.
- Similarity Checker: Compare the input and output text overlap.
- AI-Style Pattern Checker: Flag robotic phrases and overused AI-style wording.

## SEO & Blogging

- Blog Outline Generator: Create a structured article outline.
- Blog Intro Generator: Create a strong introduction.
- Blog Post Builder: Create a structured blog draft from a topic or notes.
- Content Calendar: Generate a practical 7-day content plan.
- SEO Content Brief: Generate SEO angle, headings, FAQs, and keyword suggestions.
- SEO Meta Title: Generate SEO-friendly title ideas.
- SEO Meta Description: Generate meta descriptions.
- URL Slug Generator: Create clean SEO slugs.
- Keyword Density Checker: Calculate keyword frequency.
- FAQ Builder: Generate questions and answers from content.
- FAQ Schema JSON: Create simple FAQ schema markup from your content.
- Heading Optimizer: Generate improved headings.
- Featured Snippet Writer: Create concise answer snippets.
- Search Intent Classifier: Classify informational, commercial, transactional, or navigational intent.

## Student Tools

- Study Notes Generator: Turn text into clean study notes.
- Important Points Extractor: Extract key points.
- MCQ Generator: Create multiple-choice questions.
- Flashcard Creator: Create question-answer flashcards.
- Definition Extractor: Extract terms and definitions.
- Glossary Builder: Extract terms and create simple explanations.
- Thesis Statement Generator: Create academic thesis statement options.
- Discussion Post Writer: Create a student-style discussion forum post.
- Lesson Plan Builder: Create lesson objectives, activities, and assessment ideas.
- Essay Outline Builder: Create academic essay structure.
- Citation Formatter: Format simple APA/MLA/Harvard-style references from provided details.
- Assignment Formatter: Format content into assignment structure.

- Research Summary: Summarize research-style content.
- Quiz Generator: Create short answer questions.

## **Business Tools**

- Email Writer: Draft a professional email.
- Cold Email Writer: Create a polite sales or outreach email.
- Proposal Writer: Create a business/freelance proposal.
- Meeting Notes Formatter: Turn messy notes into meeting minutes.
- Meeting Agenda Builder: Turn a topic into a clean meeting agenda.
- Customer Support Reply: Create a polite support response.
- Resume Bullet Improver: Create achievement-focused resume bullets.
- Cover Letter Builder: Draft a cover letter.
- LinkedIn Bio Generator: Create a professional profile summary.
- Job Description Writer: Create a structured job post.
- Client Brief Builder: Create a concise client/project brief.
- SWOT Analysis: Create strengths, weaknesses, opportunities, and threats.
- Product Description Rewriter: Create clean e-commerce product copy.
- Invoice Note Writer: Create professional invoice/payment notes.
- Terms / Policy Draft Helper: Draft simple website policy text.

## **Social Media**

- Viral Hook Generator: Generate attention-grabbing openings.
- YouTube Script Writer: Generate structured video script.
- Short Video Script: Create a 30-60 second script for Reels, Shorts, or TikTok.
- YouTube Title Generator: Generate title ideas.
- YouTube Description Writer: Generate a clean video description.
- TikTok Hook Writer: Generate short hooks.
- Instagram Caption Generator: Generate caption ideas.
- LinkedIn Post Writer: Create a professional LinkedIn post.
- Facebook Ad Copy: Generate headline, primary text, and CTA.
- X/Twitter Thread Writer: Create a thread outline.
- CTA Generator: Generate calls-to-action.
- Hashtag Generator: Generate hashtags.
- Carousel Post Outline: Create slide-by-slide carousel text.
- Social Media Calendar: Generate a 7-day social posting plan.

## **Document Tools**

- Clean Formatting: Remove messy spacing and improve layout.
- Markdown Converter: Convert text into clean Markdown.
- Case Formatter: Convert to title, sentence, upper, or lower case.
- Bullet Point Generator: Turn text into bullets.
- Outline Builder: Create a structured outline.
- Executive Summary: Create a concise executive summary.
- Pros & Cons Generator: Create a balanced pros and cons list.
- Timeline Extractor: Extract dates, steps, and milestones when available.



- Title Generator: Generate polished titles.
- Prompt Writer: Create a detailed prompt.
- Prompt Improver: Improve an existing prompt.
- Text to Table: Convert lines into a simple Markdown table.
- Action Items Extractor: Extract tasks, owners, and deadlines when present.
- Content Repurposer: Turn one text into posts, email, bullets, and summary.

## 7. Single Tool Workspace

Every individual text tool opens on a dedicated workspace page. The user can paste text, choose settings, run the tool, review analytics, download the result, and switch between Algorithm mode and Gemini API mode when eligible.

| Workspace Component  | Description                                                                                         |
|----------------------|-----------------------------------------------------------------------------------------------------|
| Input Area           | Large readable textarea for original text, prompt, notes, document content, or source material.     |
| Output Area          | Separate readable output panel with tinted background to distinguish final results from input text. |
| Tool Options         | Controls such as tone, strength, case format, output style, and processing mode.                    |
| Algorithm/API Toggle | Allows eligible users to switch between built-in algorithm mode and Gemini API mode.                |
| Analytics Panel      | Shows similarity, originality lift, readability, word count, and other text metrics.                |
| Export Buttons       | Download results as TXT, Markdown, DOCX, or PDF depending on available libraries.                   |
| History              | Stores recent user outputs for easier reuse.                                                        |

## 8. Algorithm Mode and Gemini API Mode

TextForge Studio supports two processing approaches. Algorithm mode works without any API. Gemini API mode uses a user-provided Google AI Studio / Gemini API key. The API key is entered by the user from the workspace settings area and is used only when the user turns on the API mode.

| Mode            | How It Works                                                                                                                                         |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Algorithm Mode  | Runs built-in Python text processing functions. No API key is required. This mode is available to free/trial/premium users according to plan limits. |
| Gemini API Mode | Uses the user's own Gemini API key for stronger AI generation. This mode is available to trial/premium users according to plan rules.                |
| Hybrid Fallback | If API mode fails, returns an error, or produces unusable output, the app falls back to built-in algorithm processing.                               |

### Important Note

The application does not need a platform owner API key to run basic tools. Users can optionally add their own Gemini API key if they want AI-assisted results.

For responsible use, the platform should be presented as a writing and originality assistant, not as a guaranteed plagiarism removal or AI detector bypass system.

## 9. Subscription and Trial System

The website includes a built-in monetization system. New users receive a 1-day trial, free users have a daily character limit, and premium users receive higher limits and premium workflows. Admin can change pricing and limits from the admin panel.

| Plan          | Description                                                                                                               |
|---------------|---------------------------------------------------------------------------------------------------------------------------|
| Free          | Basic access with limited daily characters and algorithm tools.                                                           |
| Trial         | 1-day trial with higher character limit and premium workflows available temporarily.                                      |
| Pro / Premium | Paid plan that unlocks premium tools, higher daily character limits, exports, and API mode with user-provided Gemini key. |
| Admin         | Unlimited access to all features and no daily character restrictions.                                                     |

- Default subscription price is configurable from the admin panel, starting from \$5 per month.
- Character limits are configurable for free, trial, and premium plans.
- Premium tools remain locked until the user is in trial or premium status.
- When a manual payment proof is approved by admin, the user account automatically changes to premium.

## 10. Payment Methods and Admin Approval

The final build supports both manual and checkout-link based payment workflows. International providers are configured as links/details in the admin panel. Pakistani payment methods allow users to upload screenshots for admin approval.

| Method           | Status / Behavior                                                                                      |
|------------------|--------------------------------------------------------------------------------------------------------|
| EasyPaisa        | Manual Pakistani payment proof workflow. Default number: 0340-545-3639. Account Holder: Manza Zahoor.  |
| SadaPay          | Manual Pakistani payment option. Default number: 03140895219. Account Holder: syed Muneeb Haider Shah. |
| Mashreq Neo Bank | Added as an admin-configurable bank option for later setup.                                            |
| Allied Bank      | Added as an admin-configurable bank option for later setup.                                            |
| PayPal           | Manual/display payment option controlled from admin settings.                                          |
| Payoneer         | Manual/display payment option controlled from admin                                                    |

|               |                                                                              |
|---------------|------------------------------------------------------------------------------|
|               | settings.                                                                    |
| Stripe        | Admin can paste checkout link; users are sent to the external checkout page. |
| Lemon Squeezy | Admin can paste checkout link; users are sent to the external checkout page. |
| Paddle        | Admin can paste checkout link; users are sent to the external checkout page. |
| Bank Transfer | Manual bank transfer details can be configured by admin.                     |

## 10.1 Manual Payment Proof Workflow

21. User opens the Subscribe page.
22. User chooses an enabled manual method such as EasyPaisa, SadaPay, Payoneer, PayPal, or bank transfer.
23. User sends payment to the displayed account details.
24. User uploads a screenshot or proof of payment.
25. Admin opens the Admin Panel and reviews pending proofs.
26. Admin approves or rejects the proof.
27. If approved, the user is automatically upgraded to premium.

## 11. Admin Panel Documentation

The admin panel is the control center for the website owner. It is visible only to the admin account and protected from normal users. Admin can manage monetization, AdSense, payment methods, user plans, and approvals.

| Admin Feature          | Description                                                                                                                                           |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| User Statistics        | Shows total users, premium users, trial users, and usage information.                                                                                 |
| User List              | Displays signed-up users, emails, plans, and account status.                                                                                          |
| Payment Proofs         | Shows uploaded proof files and allows approve/reject actions.                                                                                         |
| Subscription Settings  | Allows admin to change plan price, daily character limits, and plan settings.                                                                         |
| Payment Method Toggles | Allows admin to enable/disable Stripe, Lemon Squeezy, Paddle, PayPal, Payoneer, EasyPaisa, SadaPay, bank transfer, Mashreq Neo Bank, and Allied Bank. |
| AdSense Settings       | Allows admin to add global AdSense script, publisher ID, and ad blocks for header, sidebar, tool area, and footer.                                    |
| Admin Unlimited Access | Admin user can access all tools with no limits and does not need subscription approval.                                                               |

## 12. AdSense Readiness

The website includes core pages and technical endpoints that support AdSense readiness. Approval is still controlled by Google and depends on website quality, traffic, content policy compliance, and review outcome.

- Privacy Policy, Terms and Conditions, Disclaimer, About page, and guide pages are included.
- Admin can paste AdSense global script and ad placement HTML without editing code.
- Dynamic ads.txt endpoint is included through /ads.txt.
- robots.txt and sitemap.xml endpoints are included for search engine crawling.
- Ad placement areas are available for header, sidebar, tool area, and footer blocks.
- The website includes real text-tool content and feature descriptions instead of being only a blank tool page.

## 13. Export, Import, and Document Support

The platform supports multiple input and export formats. This makes the tool more useful for students, freelancers, writers, marketers, and business users who need downloadable results.

| Feature           | Description                                                                  |
|-------------------|------------------------------------------------------------------------------|
| TXT Import/Export | Plain text upload and download for quick use.                                |
| Markdown Export   | Markdown style output for notes, documentation, and content workflows.       |
| DOCX Export       | Microsoft Word compatible export through python-docx.                        |
| PDF Export        | PDF output support through ReportLab or available PDF generation utilities.  |
| PDF/DOCX Import   | Basic document text import where libraries and hosting environment allow it. |

## 14. Design and User Interface Improvements

The final version focuses on a clearer, more professional design. The default appearance is bright and calm, using light blue and white gradients. Dark mode remains available, but the website does not start in a heavy dark interface.

- Default bright theme with soft blue and white gradient backgrounds.
- Improved text contrast in cards, side panels, pricing sections, subscription pages, and text fields.
- Dedicated pages reduce messiness and make the interface easier to use.
- Tool dropdown in the main navigation allows quick category switching.
- Readable cards, better spacing, better padding, and improved mobile responsiveness.
- Workspace settings button is designed to show settings above cards instead of hiding behind content.
- Input and output areas are visually separated for easier reading and editing.
- Sidebar and plan cards are simplified to avoid clutter.

## 15. Backend Architecture

The backend is built with Flask. The text tool logic is separated into a text engine file, while routes, authentication, admin actions, payment settings, subscription logic, and page rendering are managed through the main application file.

| File/Folder           | Purpose                                                                                                                                 |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| app.py                | Main Flask application, routes, authentication, admin, subscriptions, payments, AdSense settings, and page rendering.                   |
| text_engine.py        | Built-in no-API text processing algorithms, tool categories, tool metadata, and tool execution functions.                               |
| templates/            | HTML templates for public pages, authentication, dashboard, tools, category pages, single tool pages, admin, pricing, and subscription. |
| static/style.css      | Main visual design, light/dark themes, responsive layout, cards, buttons, tool pages, and form styling.                                 |
| static/app.js         | Frontend interactivity including settings menu, theme switching, tool execution, export, and UI behavior.                               |
| instance/textforge.db | SQLite database for users, settings, subscriptions, payment proofs, and app data.                                                       |
| requirements.txt      | Python dependencies required to run the app.                                                                                            |
| Procfile / Dockerfile | Deployment support files for production hosting or container deployment.                                                                |

## 16. Database and Data Flow

The SQLite database stores user accounts, settings, payment proof data, plan status, and application configuration. It is suitable for starting and testing. For larger public traffic, PostgreSQL or MySQL is recommended.

| Flow                 | Description                                                                                         |
|----------------------|-----------------------------------------------------------------------------------------------------|
| Signup               | User completes form and CAPTCHA. Account is created and trial status begins.                        |
| Login                | User completes CAPTCHA and authenticates with email/password.                                       |
| Tool Use             | System checks plan, daily limit, and premium access before running a selected tool.                 |
| API Mode             | If enabled and allowed, the user-provided Gemini key is used. If it fails, algorithm fallback runs. |
| Subscription Request | User selects payment method, pays externally/manual, and uploads proof if needed.                   |
| Admin Approval       | Admin approves payment proof and user plan changes to premium automatically.                        |
| AdSense              | Admin saves ad script/slots, which are rendered into the configured ad areas.                       |

## 17. Deployment Guide

The project can be deployed on any hosting provider that supports Python web applications. It is not limited to Netlify, Vercel, or any single platform.

### 17.1 Local Run

- Install Python 3.10 or newer.
- Open terminal inside the project folder.
- Run: `pip install -r requirements.txt`
- Run: `python app.py`
- Open: `http://127.0.0.1:5000`

### 17.2 Production Run

- Use production command: `gunicorn app:app`
- Set environment variable `SECRET_KEY` before launch.
- Change admin password before launch.
- Use HTTPS through your hosting provider or reverse proxy.
- For large traffic, replace SQLite with PostgreSQL/MySQL and configure persistent storage for uploaded payment proof files.

### 17.3 Suggested Hosting Options

- Render, Railway, PythonAnywhere, Hostinger VPS, DigitalOcean, Hetzner, or any VPS that supports Python and Gunicorn.
- For basic public launch, use GitHub plus a Python hosting provider.
- For serious traffic and better control, use a VPS with NGINX, Gunicorn, and a process manager.

## 18. Security Checklist

### Before Public Launch

Change `ADMIN_PASSWORD` from the default value.

Set a strong `SECRET_KEY` in environment variables.

Use HTTPS on the public domain.

Do not commit real payment secrets, API secrets, or private admin credentials into GitHub.

Restrict admin access and keep payment proofs private.

Back up the database regularly.

If accepting uploaded proof images, limit file size and allowed file types.

Use secure cookies and production-safe session settings when deployed.

The current build is a strong working foundation, but any public website handling user accounts and payments should be reviewed before launch. If traffic grows, move from SQLite to a production database and use more robust authentication/security hardening.

## 19. Responsible Use and Limitations

TextForge Studio is designed as a writing assistance platform. It can rewrite, polish, structure, summarize, and analyze text, but it should not be marketed as a guaranteed plagiarism remover, guaranteed AI detector bypass, or legal/academic compliance tool.

- No system can guarantee 100% plagiarism removal without checking all online, offline, academic, and copyrighted sources.
- Users should cite sources when borrowing ideas, facts, or research.
- Algorithm mode improves wording and structure, but human review is still needed.
- Gemini API mode depends on the user's own API key and Google API behavior.
- Outputs should be reviewed for accuracy, grammar, tone, and originality before publishing or submitting.

## 20. Future Upgrade Ideas

The final build already includes many tools and monetization features. Future development can turn it into a larger SaaS platform.

- Add real card subscription automation using Stripe, Lemon Squeezy, or Paddle webhooks.
- Add email verification and password reset workflow.
- Add full user profile page and invoice history.
- Add admin analytics charts for revenue, active users, tool usage, and conversion rate.
- Add team/workspace plans for agencies and schools.
- Add PostgreSQL support for production scalability.
- Add background jobs for long document processing.
- Add better local AI models on a VPS/GPU server if no external API is desired.
- Add multilingual tools for Urdu, Hindi, English, and other languages.
- Add a cleaner SaaS pricing/paywall system with invoices and automatic renewals.

## Final Notes

TextForge Studio is now a platform-ready text tools website with professional pages, legal pages, monetization, admin management, payment proof approvals, AdSense controls, and multi-page tool workflows. It is suitable as a digital product, SaaS starter, or hosted writing assistant platform.

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